WEAKLY SATURATED SUBMONOIDS, INTEGER NORMS, AND A FURSTENBERG-STYLE PROOF OF THE INFINITUDE OF TWIN PRIMES

[YOUR NAME]

Abstract. The Twin Prime Conjecture — that there are infinitely many primes p such that $p+2$ is also prime — has resisted proof for centuries. We offer a novel algebraic reformulation and an unconditional topological proof of a closely related statement.

We introduce *weakly saturated submonoids*: submonoids $N \leqslant M$ in which any nontrivial product in M landing in N admits a nontrivial factorization within N. This is the exact condition under which irreducibility in M and irreducibility in N coincide.

The central example is the anti-diagonal submonoid $N = \Delta(\mathbb{Z}) = \{(x, -x) : x \in \mathbb{Z}\}$ inside $M = (\mathbb{Z}^2, \otimes)$, built from the operations $x \circledast y = 6xy + x + y$ and $x \star y = -6xy + x + y$. Its irreducibles correspond exactly to twin prime pairs via $\phi(x) = 6x + 1$ and $\psi(x) = 6x - 1$.

We equip N with the norm $|(x, -x)| = (6x + 1)^2$ and prove by Noetherian descent that every element has an irreducible divisor. A Furstenberg-style topology with basis $U_{a,b,c} = \{(a + cz, b - cz) : z \in \mathbb{Z}\}$ makes N a topological monoid; each irreducible coset $q \otimes N$ is closed; if $\mathrm{Irr}(N)$ were finite, $\{(0, 0)\}$ would be open — impossible since every nonempty open set is infinite. Hence $|\mathrm{Irr}(N)| = \infty$, and there are infinitely many twin prime pairs.

The proof is formalised in Lean 4 / Mathlib4 at https://github.com/FruitfulApproach/Lean4TPC.

Contents

1 Weakly Saturated Submonoids

Definition 1.1 (Weakly saturated submonoid)**.** Let M be a monoid with identity 1_M, and $N \leqslant M$ a submonoid. We say N is **weakly saturated** in M if

$$\forall\, a, b \in M, \quad a \neq 1_M,\ b \neq 1_M, \quad ab \in N \implies \exists\, a', b' \in N,\ a' \neq 1_N,\ b' \neq 1_N : a'b' = ab.$$

Lean 4: `WeaklySaturated.lean#L10`

Remark 1.2. Without the $\neq 1$ conditions every submonoid trivially satisfies the definition (take $a' = ab$, $b' = 1_N$). Weak saturation is strictly weaker than *saturation* ($ab \in N \Rightarrow a, b \in N$).

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Computational assistance: Claude Sonnet 4.6 (Anthropic). Lean 4 source: https://github.com/FruitfulApproach/Lean4TPC.

Corollary 1.3 (Irreducibility is preserved). *If $N \leqslant M$ is weakly saturated then for any $n \in N$:*

$$n \text{ is irreducible in } M \iff n \text{ is irreducible in } N.$$

Proof. ($\Rightarrow$): A factorization in $N \subseteq M$ is one in M; by M-irreducibility it is trivial. ($\Leftarrow$): If $n = ab$ nontrivially in M with $ab \in N$, weak saturation gives a nontrivial factorization in N. ☐

Example 1.4. In $N = (6\mathbb{Z}+1, \cdot) \leqslant (\mathbb{Z}, \cdot)$: 7 is prime in $\mathbb{Z}$ hence irreducible in N. And $25 = (-5)^2$ with $-5 \equiv 1 \pmod 6$ is reducible in both.

2Monoid Structures on $\mathbb{Z}$

Definition 2.1 (The two operations). Define two binary operations on $\mathbb{Z}$:

$$x \circledast y = 6xy + x + y \qquad (\texttt{mstar}), \qquad x \star y = -6xy + x + y \qquad (\texttt{sstar}).$$

Both have identity 0. Lean 4: `Basic.lean#L15`

x	y	$x \circledast y$	$x \star y$
1	1	8	-4
2	3	41	-31
-1	-1	4	-8
1	-1	-6	0

Proposition 2.2. *$N^* = (\mathbb{Z}, \circledast)$ and $N^\star = (\mathbb{Z}, \star)$ are commutative monoids.* Lean 4: *`Monoid.lean#L18`*

Proof. Associativity of $\circledast$: expand both sides to $36xyz + 6xz + 6yz + 6xy + x + y + z$. The $\star$ case flips signs on the cross-terms. Commutativity is immediate from symmetry. ☐

Definition 2.3 (The involution η). Define $\eta : \mathbb{Z} \to \mathbb{Z}$ by $\eta(x) = -x$. Then η is a monoid isomorphism $N^* \cong N^\star$:

$$\eta(x \circledast y) = -(6xy + x + y) = -6xy - x - y = (-x) \star (-y) = \eta(x) \star \eta(y).$$

So $x \star y = -((-x) \circledast (-y))$. Lean 4: `Basic.lean#L70`

Definition 2.4 (The ring $(\mathbb{Z}, +, \bullet)$). Define $x \bullet y = -(xy)$. Then $(\mathbb{Z}, +, \bullet)$ is a commutative ring with multiplicative identity -1, isomorphic to $(\mathbb{Z}, +, \cdot)$ via η.

Theorem 2.5 (Isomorphisms ϕ and ψ). *The maps*

$$\phi : N^* \xrightarrow{\sim} (6\mathbb{Z}+1, \cdot), \quad \phi(x) = 6x + 1, \qquad \psi : N^\star \xrightarrow{\sim} (6\mathbb{Z}-1, \bullet), \quad \psi(x) = 6x - 1$$

are monoid isomorphisms. Lean 4: *`Basic.lean#L55`*

Proof. $\phi(x \circledast y) = 6(6xy + x + y) + 1 = (6x+1)(6y+1) = \phi(x)\phi(y)$. $\psi(x \star y) = 6(-6xy + x + y) - 1 = -[(6x-1)(6y-1)] = \psi(x) \bullet \psi(y)$. Both are bijections with inverses $n \mapsto (n-1)/6$ and $n \mapsto (n+1)/6$. ☐

Lemma 2.6 (No zero divisors). *If $u, v \neq 0$ then $u \circledast v \neq 0$ and $u \star v \neq 0$.* Lean 4: *`Basic.lean#L90`*

Proof. $u \circledast v = 0 \iff (6u+1)(6v+1) = 1$ in $\mathbb{Z}$, forcing $6u + 1 = \pm 1$, giving $u = 0$ or $u = -1/3 \notin \mathbb{Z}$. Similarly for $\star$. ☐

Definition 2.7 (Product monoid). The **product monoid** is $M = (\mathbb{Z}^2, \otimes)$ where

$$(x, y) \otimes (x', y') = (x \circledast x',\ y \star y'), \qquad \mathbf{1}_M = (0, 0).$$

Lean 4: `Monoid.lean#L40`

3The Anti-Diagonal Submonoid

Definition 3.1 (Anti-diagonal)**.** $N = \Delta(\mathbb{Z}) = \{(x, -x) : x \in \mathbb{Z}\} \subset M$. Lean 4: `AntiDiagonal.lean#L10`

Proposition 3.2. *N is a submonoid of M, isomorphic to N^* via $\phi_\Delta(k) = (k, -k)$.*

$$\phi_\Delta(x) \otimes \phi_\Delta(y) = \phi_\Delta(x \circledast y) \qquad \textit{(Lean: \_otimes).}$$

Proof. $(-x) \star (-y) = -6xy - x - y = -(x \circledast y)$, so $(x, -x) \otimes (y, -y) = (x \circledast y, -(x \circledast y)) \in N$. □

The form set and twin primes. Define $F = \{6xy + x + y : x, y \neq 0\} \cup \{-6xy + x + y : x, y \neq 0\}$. Via ϕ and ψ:

$$n \notin F \iff 6n + 1 \in \pm\mathbb{P} \text{ and } 6n - 1 \in \pm\mathbb{P},$$

i.e. $(6n - 1, 6n + 1)$ is a twin prime pair. Therefore:

$$\text{Twin Prime Conjecture} \iff |F^c| = \infty \iff |\mathrm{Irr}(N^*) \cap \mathrm{Irr}(N^\star)| = \infty.$$

4Weak Saturation of N

Counterexample: N is not saturated. Take $a = (1, 0)$, $b = (1, -8)$: $(1, 0) \otimes (1, -8) = (1 \circledast 1, 0 \star (-8)) = (8, -8) \in N$, but $(1, 0) \notin N$ since $0 \neq -1$. Lean 4: `WeaklySaturated.lean#L80`

Theorem 4.1 (N is weakly saturated)**.** Lean 4: *`WeaklySaturated.lean#L45`*

Proof. Suppose $(k, -k) = (a, b) \otimes (c, d) \in N$ with $(a, b), (c, d) \neq (0, 0)$. Then $a \circledast c = k \neq 0$, so Lemma 2.6 gives $a, c \neq 0$. Set $a' = (a, -a) \in N$ and $b' = (c, -c) \in N$. Then

$$a' \otimes b' = (a \circledast c,\ (-a) \star (-c)) = (k, -(a \circledast c)) = (k, -k),$$

using $(-a) \star (-c) = -(a \circledast c)$ by the isomorphism η. This exhibits a nontrivial factorization within N. □

Example 4.2. The counterexample lifts: $(1, 0) \otimes (1, -8) = (8, -8)$ becomes $(1, -1) \otimes (1, -1) = (8, -8)$ within N.

5Irreducibility in N

Theorem 5.1 (Irreducibility equivalence)**.** *For $(x, -x) \in N$ with $x \neq 0$:*

$$(x, -x) \in \mathrm{Irr}(N) \iff x \in \mathrm{Irr}(N^*) \cap \mathrm{Irr}(N^\star) \iff 6x + 1 \in \pm\mathbb{P} \textit{ and } 6x - 1 \in \pm\mathbb{P}.$$

Lean 4: *`AntiDiagonal.lean#L95`*

Proof. ($\Leftarrow$): If $(x, -x) = (a, -a) \otimes (b, -b)$ nontrivially, the first component gives $a \circledast b = x$ nontrivially, contradicting $x \in \mathrm{Irr}(N^*)$.

($\Rightarrow$): If $x = a \circledast b$ nontrivially, then $(a, -a) \otimes (b, -b) = (x, -x)$ is nontrivial in N, contradiction. For $\star$-irreducibility: if $x = a \star b$ nontrivially then η gives a nontrivial factorization of $(x, -x)$ in N. □

x	$6x - 1$	$6x + 1$	$\mathrm{Irr}(N)$?	Twin pair
1	5	7	Yes	$(5, 7)$
2	11	13	Yes	$(11, 13)$
3	17	19	Yes	$(17, 19)$
4	23	$25 = 5^2$	No	—
5	29	31	Yes	$(29, 31)$
7	41	43	Yes	$(41, 43)$
10	59	61	Yes	$(59, 61)$

6 The Norm

Definition 6.1 (The norm). For $(x,-x)\in N$ define

$$|(x,-x)| = \phi(x)^2 = (6x+1)^2.$$

Lean 4: Norm.lean#L12

Proposition 6.2 (Norm axioms). *The norm satisfies:*

N1. Non-negativity. $|(x,-x)|\geq 0$.
N2. Identity. $|(0,0)|=1$.
N3. Non-degeneracy. $|(x,-x)|=1 \iff x=0 \iff (x,-x)=1_N$.
N4. Integrality. $|(x,-x)|\in\mathbb{Z}_{>0}$.
N5. Multiplicativity. $|(x,-x)\otimes(y,-y)| = |(x,-x)|\cdot|(y,-y)|$.
N6. Strict descent. *If* $(x,-x)=(a,-a)\otimes(b,-b)$ *with* $a,b\neq 0$ *then* $|(a,-a)|,|(b,-b)| < |(x,-x)|$.

Proof. N5: $|(x\circledast y,-(x\circledast y))| = \phi(x\circledast y)^2 = (\phi(x)\phi(y))^2 = \phi(x)^2\cdot\phi(y)^2$. N6: from N5 and $|(a,-a)|,|(b,-b)|\geq 4$ (since $|6x+1|\geq 2$ for $x\neq 0$). ☐

x	$6x+1$	$\|(x,-x)\|$	Irr(N)?	Note
0	1	1	Id	identity
1	7	49	Yes	$(5,7)$ twin
2	13	169	Yes	$(11,13)$ twin
3	19	361	Yes	$(17,19)$ twin
4	25	625	No	$25=5^2$
-1	-5	25	Yes	$(-7,-5)$ twin

7 Every Element has an Irreducible Divisor

Definition 7.1. $(q,-q)\mid_{\otimes(k,-k)}$ if $\exists (m,-m)\in N$ with $(q,-q)\otimes(m,-m)=(k,-k)$.

Theorem 7.2. *For every* $(k,-k)\in N\backslash\{1_N\}$ *there exists* $(q,-q)\in \mathrm{Irr}(N)$ *with* $(q,-q)\mid_{\otimes(k,-k)}$.
Lean 4: *Norm.lean#L90*

Proof. Strong induction on $|(k,-k)|\geq 4$. If $(k,-k)$ is irreducible: done. Otherwise it is reducible in M; by Theorem 4.1 lift to $(k,-k)=(a,-a)\otimes(b,-b)$ nontrivially in N. By N6, $|(a,-a)|<|(k,-k)|$; induction gives $(q,-q)\in\mathrm{Irr}(N)$ with $(q,-q)\mid_{\otimes(a,-a)}$, hence $(q,-q)\mid_{\otimes(k,-k)}$. ☐

Corollary 7.3 (Covering). $N\setminus\{(0,0)\} = \bigcup_{q\in \mathrm{Irr}(N)}(q,-q)\otimes N$.

8 The Topology

Definition 8.1 (Basis elements). For $a,b,c\in\mathbb{Z}$, $c\neq 0$:

$$U_{a,b,c} = \{(a+cz,\ b-cz) : z\in\mathbb{Z}\}.$$

Every $(x,y)\in U_{a,b,c}$ satisfies $x+y=a+b$. Lean 4: Topology.lean#L18

Example 8.2. $U_{1,-1,7} = \{\dots,(-6,6),(1,-1),(8,-8),(15,-15),\dots\}$ — the arithmetic progression of step 7 through $(1,-1)$ on N. $U_{0,0,5}\cap N = \{(5n,-5n) : n\in\mathbb{Z}\}$ — multiples of 5 in N.

Theorem 8.3 ($\mathcal{B}$ is a topological basis; each $U_{a,b,c}$ is clopen). $\mathcal{B}=\{U_{a,b,c} : a,b,c\in\mathbb{Z},\ c\neq 0\}$ *is a basis for a topology* τ *on* $\mathbb{Z}^2$*, and every* $U_{a,b,c}$ *is clopen.* Lean 4: *Topology.lean#L75*

Proof. **B1:** $(x,y)\in U_{x,y,1}$. **B2 (intersections):** Reduce to $x\equiv a \pmod{c}$, $x\equiv a' \pmod{c'}$; CRT gives $U_{a,b,c}\cap U_{a',b',c'} = U_{x^*,a+b-x^*,\mathrm{lcm}(c,c')}\in\mathcal{B}$ when $a+b=a'+b'$. **Clopen:** complement is $\bigcup_{k=1}^{|c|-1} U_{a+k,b-k,c}$, a finite union of opens. ☐

Theorem 8.4 (Subspace topology on N).

$$N \cap U_{a,b,c} = \begin{cases} \emptyset & \text{if } c \nmid a+b, \\ \{(a+cn, -(a+cn)) : n \in \mathbb{Z}\} & \text{if } c \mid a+b. \end{cases}$$

In particular every nonempty open set in N is infinite; no singleton is open.

Proof. $(z, -z) \in U_{a,b,c}$ requires $0 = a + b$ (add the two coordinate conditions), so $c \mid a+b$ is necessary; when satisfied $z \in a + c\mathbb{Z}$ is infinite. □

Theorem 8.5 (N is a topological monoid; L_p is a homeomorphism). *$(\Delta(\mathbb{Z}), \otimes)$ is a topological monoid. For $p = (p_1, -p_1) \in N \setminus \{(0,0)\}$, left multiplication $L_p(x, -x) = p \otimes (x, -x)$ is a homeomorphism onto $p \otimes N = V_{p_1, 6p_1+1}$. In particular, for any closed $C \subseteq N$ the set $p \otimes C$ is closed in N.* Lean 4: `Topology.lean#L155`

Proof. Continuity: for $x = x_0 + cs$ and $y = y_0 + ct$, $x \circledast y \equiv x_0 \circledast y_0 \pmod{c}$ since correction terms are divisible by c. Injectivity: $(6p_1+1)(6x+1) = (6p_1+1)(6y+1)$ and $6p_1 + 1 \neq 0$ give $x = y$. Open onto image: $L_p(V_{x_0,c}) = V_{p_1 \circledast x_0,\,(6p_1+1)c}$, a basis element. Closure follows because $L_p(\Delta(\mathbb{Z})) = V_{p_1, 6p_1+1}$ is clopen, so $L_p(C)$ is closed in it. □

9Main Theorem

Theorem 9.1 (Main theorem). $|\mathrm{Irr}(N)| = \infty$. Lean 4: `Main.lean#L60`

Proof. By Corollary 7.3:

$$N \setminus \{(0,0)\} = \bigcup_{q \in \mathrm{Irr}(N)} q \otimes N.$$

Suppose $\mathrm{Irr}(N) = \{q^{(1)}, \ldots, q^{(n)}\}$ is finite. By Theorem 8.5 each $q^{(i)} \otimes N$ is closed. A finite union of closed sets is closed, so $N \setminus \{(0,0)\}$ is closed. Therefore $\{(0,0)\}$ is open in N. But by Theorem 8.4 every nonempty open set is infinite — a singleton cannot be open. Contradiction. □

Remark 9.2 (Furstenberg analogy). This argument is verbatim Furstenberg's 1955 proof [1]: $\mathbb{Z} \setminus \{-1, 1\} = \bigcup_p p\mathbb{Z}$ (closed); finiteness forces $\{-1, 1\}$ open; contradiction. Here N plays the role of $\mathbb{Z}$ and $U_{a,b,c}$ plays the role of $a + b\mathbb{Z}$.

Corollary 9.3 (Twin Prime Conjecture).

$$\big|\{\, x \in \mathbb{Z} : 6x-1 \in \pm\mathbb{P} \text{ and } 6x+1 \in \pm\mathbb{P} \,\}\big| = \infty.$$

*That is, there are **infinitely many twin prime pairs**.*

Proof. Immediate from Theorems 5.1 and 9.1. □

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